

Research & Requirements Document

Intelligent Medication Dispenser for Compliance

1. Introduction

Missed doses, wrong-pill mistakes, and medication degradation from poor storage conditions are common causes of treatment failure in patients managing multiple prescriptions. This project develops a compact, ESP32-based bedside device that automates dispensing, verifies each dose physically, monitors storage conditions, and reports compliance data to a caregiver or clinical dashboard.

This document summarizes background research on the existing market, restates the project's functional and non-functional requirements, and defines the engineering assumptions and constraints that will guide the design phase.

2. Problem Statement

A patient on a complex regimen (e.g. five medications at different times of day) is at risk of:

- Missing a dose or taking it late
- Taking the wrong pill
- Medication degrading due to heat or humidity (e.g. in a bathroom environment)
- These failures going unnoticed until they cause a health setback

The device addresses this by acting as a local, intelligent hub: it physically secures medication, dispenses the correct dose at the correct time, verifies the action mechanically, and escalates to alerts and logging when something goes wrong.

3. Background Research

3.1 Market Context

The smart pill dispenser market is a fast-growing segment of digital health, driven by an aging population, rising chronic disease rates, and expansion of home healthcare and remote patient monitoring. Multiple industry analyses place the global market in the low-to-mid single-digit billions of dollars as of 2026, with continued double-digit annual growth projected through the early 2030s, driven in large part by Bluetooth/Wi-Fi connected devices and integration with caregiver-facing apps.

3.2 Existing Products (Related Work)

Product	Approach	Notable strengths	Notable limitations
Hero	Robot-style dispenser; app-managed schedule; button-press dispensing	Holds up to ~90 days / 10 medications; strong app + missed-dose alerts to a "care circle"	Cannot handle gel caps, gummies, or half pills; requires home Wi-Fi
MedMinder	Pharmacy pre-sorts medication into a tray; touchscreen interface	Very easy setup; built-in cellular option (no Wi-Fi needed); individual compartment locking	Subscription-based (recurring cost); relies on pharmacy for sorting
MedaCube	Self-contained robot organizer with barcode scanner	Barcode scanning simplifies data entry; large single-button interface aids accessibility	Higher one-time cost; same pill-type limitations as Hero
MedReady / Philips Lifeline / TabSafe / GMS	Simpler locked rotating trays	Tamper-resistant, cellular alert options, lower cost tiers	Manual pill sorting by caregiver; fewer smart features

Takeaways for this project:

- Nearly every commercial product **verifies compliance through app/user interaction, not physical sensing of the pill itself** — this project's optical (IR break-beam) verification step is a meaningful differentiator, directly detecting whether a pill physically left the carousel.
- **Environmental protection (temperature/humidity) is largely absent** from consumer products reviewed — most focus on scheduling and locking, not storage-condition integrity. This is a specific gap this project's "Chamber Fault" logic addresses.

- **Connectivity approach varies** (Wi-Fi vs. cellular) — for this prototype, Wi-Fi telemetry to a monitoring node is the practical choice given the six-week scope and ESP32 hardware.

3.3 Component/Technique Research

- **IR break-beam sensors** are a standard low-cost method for detecting object pass-through in a chute, well suited to confirming a pill has dropped without needing a camera or load cell.
- **DHT22 (or similar) sensors** provide combined temperature/humidity readings on a single digital pin, appropriate for the chamber-condition monitoring requirement.
- **DS3231-class RTC modules with coin-cell backup** are the standard way to preserve schedule timing across power loss/brownouts on microcontroller projects, directly matching the brief's power-resilience assumption.
- **Stepper motors with basic drivers** (e.g. 28BYJ-48/ULN2003 for low torque, or NEMA-17/A4988 for more robust carousels) are commonly used for indexed, repeatable rotation — appropriate for a carousel that must stop at precise pill positions.

4. Functional Requirements

ID	Requirement
FR-1	The device shall store multiple medication types in physically separated carousel compartments.
FR-2	The device shall dispense the correct medication and dose at pre-programmed times.
FR-3	The device shall verify, via an optical break-beam gate, whether a pill physically dropped after a dispense action.
FR-4	If no pill is detected, the device shall execute an anti-jam retry (shake) routine before logging a dispensing fault.
FR-5	The device shall continuously monitor internal chamber temperature and relative humidity.
FR-6	If temperature $\geq 25^{\circ}\text{C}$ or relative humidity $\geq 60\%$ persists for more than 30 minutes, the device shall raise a "Chamber Fault."

FR-7	The device shall alert the patient locally (OLED + buzzer) at dose time and on any fault condition.
FR-8	The device shall log all compliance events (on-time, missed, fault) with timestamps.
FR-9	The device shall transmit status, environmental readings, and fault/compliance events to a central monitoring node over a network interface.
FR-10	The device shall retain schedule and timekeeping across a power interruption.

5. Non-Functional Requirements

ID	Requirement
NFR-1	Reliability — the anti-jam and retry logic should recover from the majority of mechanical misfires without human intervention.
NFR-2	Timeliness — dose reminders and fault alerts should trigger within a small, defined tolerance of the scheduled/threshold time (e.g. within seconds, not minutes).
NFR-3	Usability — local alerts (OLED text + buzzer pattern) should be understandable without a companion app, given the target user may be elderly or have cognitive load concerns.
NFR-4	Data integrity — logged events must include an accurate timestamp even after a power interruption, per the RTC/backup-battery assumption.
NFR-5	Maintainability — firmware should be structured (e.g. as an explicit finite state machine) so states and fault paths can be modified or extended without a full rewrite.
NFR-6	Cost/scope fit — component choices should stay within a hobbyist/prototype budget and be buildable within the six-week program timeline.

6. Engineering Assumptions (from project brief)

1. **Environmental sensitivity & segregation** — the dispenser is assumed to handle moisture/heat-sensitive medications (e.g. effervescent tablets, soft gel capsules). Chamber must stay below 60% RH and 25°C; breaches sustained over 30 minutes trigger a Chamber Fault.

2. **Mechanical tolerance & miscalculation compensation** — friction variation, pill dust, or dimensional irregularities can stall the stepper or misalign the carousel. Firmware must assume this margin of error and use the IR break-beam gate plus an anti-jam shake routine before logging a dispensing error.
 3. **Power resilience & timekeeping** — the device must never lose schedule tracking due to power fluctuation or accidental unplugging. This assumes an independent RTC module with a dedicated coin-cell backup.
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7. Scope Boundaries (for the six-week prototype)

In scope:

- Single-patient, single-unit prototype (not a multi-unit or multi-patient system)
- Core dispensing cycle, IR verification, jam retry, environmental monitoring, local alerts, logging, and basic network telemetry
- A defined firmware FSM covering the states above

Likely out of scope for a six-week build (flag for stretch goals only):

- Companion mobile app (a simple dashboard/serial log may substitute)
 - Pharmacy integration or barcode-based pill identification
 - Tamper detection / physical security locking
 - Support for non-solid dosage forms (liquids, gels, half-pills)
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8. Open Questions to Resolve Before Design Sign-off

- How many carousel compartments / medication types must the prototype support (project brief mentions five in the scenario — is that the target)?
 - What communication method for telemetry — Wi-Fi/MQTT to a cloud endpoint, or a simpler local dashboard for demo purposes?
 - What counts as "patient took the dose" — is dispensing itself treated as compliance, or is there a separate confirmation step?
 - What is an acceptable number of anti-jam retry attempts before declaring a dispensing fault?
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9. Future Work — Scaling to an Institutional Vending-Style Distributor

The long-term vision for this project is to scale beyond a single-patient bedside unit into a large, multi-medication dispensing station — closer to a pharmaceutical vending machine deployed in a pharmacy, clinic, or care facility. This section captures how that vision differs from the current six-week prototype, for future reference rather than immediate implementation.

How the scaled version would differ:

Aspect	Current prototype	Scaled vending-style distributor
Users	Single patient	Many patients/staff, likely with authentication (PIN, card, or ID scan)
Capacity	A handful of carousel slots	Many independent medication cartridges/lanes, vending-machine style
Dispensing mechanism	Single stepper-driven carousel	Multiple parallel lanes or a robotic picking arm, closer to industrial vending hardware
Verification	Single IR break-beam gate	Per-lane sensing, possibly combined with weight/load-cell or vision-based pill counting
Identity/records	None (single known patient)	Integration with patient records or prescription IDs to prevent dispensing errors
Environmental monitoring	Single chamber	Possibly zoned monitoring, since different medications may need different storage conditions
Connectivity	Single device to one monitoring node	Fleet-level telemetry — many units reporting to a central pharmacy/clinical system
Regulatory considerations	Out of scope for a prototype	Would need to account for pharmacy dispensing regulations, controlled-substance handling, and audit trails

Business model	N/A	Likely institutional purchase or lease, similar to existing commercial players (Hero, MedMinder, MedaCube) but positioned for clinics/pharmacies rather than individual homes
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Why this matters now: even though the six-week prototype won't implement any of this, keeping the current firmware modular (the FSM approach, and separating sensing/control/actuation) makes it easier to extend later — e.g. swapping a single carousel state for a "select lane N" parameter, or scaling the telemetry layer to handle multiple device IDs instead of one.

Suggested next steps, if this direction is pursued later:

- Research industrial vending/dispensing mechanisms (lane-based vending hardware, robotic pick-and-place) as a mechanical reference point instead of a single carousel.
- Investigate what regulatory/compliance requirements apply to automated pharmaceutical dispensing at an institutional scale.
- Revisit the FSM and telemetry design with multi-unit, multi-user identity in mind before scaling the codebase.

10. References

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